

Progress Test 1.A — Elec-S1

1.A · From potential to electric voltage

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 7 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion

(1 mark)

Convert the voltage $u = 0.0258$ MV into millivolts. Give the exact result in scientific notation, without rounding.

$u = 0.0258 \times 10^6 \text{ V} = 2.58 \times 10^4 \text{ V} = 2.58 \times 10^4 \times 10^3 \text{ mV}$
$u = 2.58 \times 10^7 \text{ mV}$

Marking

①
0 1
☐ ☐

Marking scheme

① 1 pt 2.58×10^7 mV, exact, with the unit symbol written correctly.

Examiner's tip

M (mega, 10^6) and m (milli, 10^{-3}) differ only by their case, exactly like k (kilo) and K (kelvin) in Test 1. Go through the base unit: MV $\rightarrow$ V $\rightarrow$ mV.

Exercise 2. Electric voltage — definition

(5 marks)

Define the electric voltage u_{AB} between two points A and B of a circuit. Write down its expression in terms of the electric potentials, deduce the relation between u_{AB} and u_{BA} , and state the SI unit of voltage (name and symbol).

The voltage u_{AB} is the difference of electric potential between A and B. Only potential differences can be measured (like a difference in height: a height needs a reference, a height difference does not). $u_{AB} = V_A - V_B$ hence $u_{BA} = V_B - V_A = -(V_A - V_B) = -u_{AB}$. Voltage is a signed quantity: $u_{AB} > 0 \iff V_A > V_B$. SI unit: the volt, symbol V (same unit as the potential).

Marking

①
0 1 2
☐ ☐ ☐

②
0 1 2
☐ ☐ ☐

③
0 1
☐ ☐

Marking scheme

- ① **2 pts** Voltage = potential difference between the two points (1 pt only if the potentials are not mentioned).
- ② **2 pts** $u_{AB} = V_A - V_B$ (1 pt) and $u_{BA} = -u_{AB}$ (1 pt).
- ③ **1 pt** Name *and* symbol: volt, V.

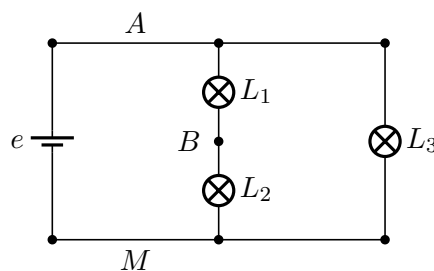
Examiner's tip

In Test 1 the unit mark was lost on a symbol. Always give both the name and the exact symbol of a unit: "volt (V)".

Exercise 3. Potentials in a circuit

(3 marks)

The circuit below contains an ideal voltage source of electromotive force $e = 6.0\text{ V}$ and three lamps. The wires are ideal. The wire M , connected to the $-$ terminal of the source, is taken as the reference of potentials: $V_M = 0$.



How many distinct electric potentials are there in this circuit?

- ☐ 2 ☒ 3 ☐ 4 ☐ 7

Why: an ideal wire has the same potential at every point, so count the electrically distinct wires: A (top), B (between L_1 and L_2) and M (bottom). The dots along one wire do not create new potentials.

The voltage across L_2 is $u_{BM} = 2.0\text{ V}$. The voltage u_{AB} across L_1 is:

- ☐ 2.0 V ☒ 4.0 V ☐ 6.0 V ☐ 8.0 V

Why: $V_A = e = 6.0\text{ V}$ and $V_B = u_{BM} = 2.0\text{ V}$, so $u_{AB} = V_A - V_B = 4.0\text{ V}$.

A voltmeter has its V terminal on B and its COM terminal on A . It displays:

- ☐ +4.0 V ☒ -4.0 V ☐ +2.0 V ☐ -6.0 V

Why: a voltmeter displays $V_V - V_{\text{COM}} = V_B - V_A = u_{BA} = -u_{AB}$. The sign is negative because COM is not on the side of the $-$ terminal.

(5 marks)

$V_A = 12.0 \text{ V}$	$V_B = 4.5 \text{ V}$	$V_C = -3.0 \text{ V}$
$u_{AB} = V_A - V_B = 12.0 \text{ V} - 4.5 \text{ V} = 7.5 \text{ V}$ $u_{BC} = V_B - V_C = 4.5 \text{ V} - (-3.0 \text{ V}) = 7.5 \text{ V}$ $u_{CA} = V_C - V_A = -3.0 \text{ V} - 12.0 \text{ V} = -15.0 \text{ V}$ $u_{AB} + u_{BC} + u_{CA} = (V_A - V_B) + (V_B - V_C) + (V_C - V_A) = 0$ (check: $7.5 \text{ V} + 7.5 \text{ V} - 15.0 \text{ V} = 0$). Going round the closed path $A \rightarrow B \rightarrow C \rightarrow A$ brings you back to the starting potential, like walking round a mountain back to your starting height. The result does not depend on the reference chosen. (This is the mesh law of section 3.A.)		

1

01

2

01

3

01

4

01

5

01

- ① **1 pt** Literal expressions $u_{XY} = V_X - V_Y$ written before any number.
- ② **1 pt** $u_{AB} = 7.5 \text{ V}$.
- ③ **1 pt** $u_{BC} = 7.5 \text{ V}$ (negative potential handled correctly).
- ④ **1 pt** $u_{CA} = -15.0 \text{ V}$ (the sign is part of the answer).
- ⑤ **1 pt** Sum = 0, justified (potentials cancel two by two / closed path).

Test 1, Ex. 4: “Never mix figures and letters in the same equation.” Write the literal relation first, then substitute *with units*. Brackets around a negative value, $4.5\text{ V} - (-3.0\text{ V})$, prevent the classic sign slip.

Unofficial practice material written from the Elec-St Part 1 lecture notes · not an official examination paper